Crop Stress Detection: Presentation & Viva Guide

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This guide has three parts: **(1)** what to say on each of the 14 slides (about 11 minutes in total), **(2)** the key terms in plain English, and **(3)** the questions the panel is likely to ask, with short answers you can say out loud. A numbers sheet is on the last page.

Green = say it | Blue = what it is | Amber = trap or follow-up | Rose = don't say this | ★ = must know

YOUR 30-SECOND PITCH (LEARN THIS BY HEART)

"My project detects crop stress from Sentinel-2 satellite images **without any labels**. I worked on 3,328 farmland tiles in Lombardy, Italy, for the 2019 season. The system uses Local Outlier Factor, a threshold set separately for each tile, and an alarm rule that needs two passes in a row. On tiles it never saw, it reaches **precision 0.876, recall 0.814 and F1 0.844**. Two findings stood out: the hybrid of three models I started with was actually worse than LOF alone, and the biggest improvement came from **where the threshold is computed**, not from a fancier model."

HOW TO STUDY THIS IN ABOUT AN HOUR

- **20 min:** read Part 1 aloud twice, with the slides open.
- **10 min:** learn the numbers sheet (last page). Cover the right column and test yourself.
- **20 min:** go through every ★ question in Part 3 and answer it out loud before reading the answer.
- **10 min:** read the amber trap questions and the "don't say" list.

Part 1 · What to say on each slide

Keep it conversational. You don't need to read the slide; say these lines and point at the picture. The orange line is the one sentence to land on each slide.

Slide 1 · Title

~25 s

"Good morning. I'm Prakhar Paawan, and with my teammate Pratham Sharma, from the Department of Information Technology, our PBL project under the guidance of Ms. Nandani Sharma is unsupervised crop stress detection from Sentinel-2 satellite data. In one line: we built a system that finds stressed farmland in satellite images without using any labels. We'll cover the problem, how the system works, how we tested it, the results, and the research paper that came out of it."

Slide 2 · The problem

~55 s

"Crop stress, from drought, disease or pests, cuts yield, often before a farmer can see it. Catching it early gives farmers time to act: to irrigate, to spray, or to send someone to check. Sentinel-2 gives free images of every field about every five days, so the data is there. The problem is labels: a normal supervised model needs experts to visit fields and record which were stressed, and that can't scale to a whole region. So I treated stress as an anomaly, something unusual compared with a tile's normal behaviour, and used no labels at all. That means the method can be used anywhere satellite images exist."

Key line: the bottleneck is labels, not images.

Slide 3 · Objectives ~40 s

"I had five goals. First, build a pipeline that cleans the satellite images and turns them into numbers. Second, compare three different detectors and a hybrid that combines them, to find which works best. Third, design the rule that decides when to actually raise an alarm. Fourth, test it honestly, even though there is no ground truth. And fifth, make it reproducible, so one command rebuilds every result."

Slide 4 · Data ~55 s

"The data is 3,328 tiles of farmland in Lombardy, northern Italy, a heavily farmed region with maize, rice and wheat. Each tile is 480 by 480 metres, so one tile usually covers a few fields. I used the 2019 season, and after removing cloudy images I kept 25 passes per tile, which gives 83,200 observations. I used MSAVI, a vegetation index like NDVI that corrects for bare soil early in the season. Each observation becomes 16 numbers: the median, the spread, percentiles, and how much it changed from earlier passes. The picture shows one tile across the year; you can see the vegetation rise and fall with the season."

Slide 5 · How the system works ~55 s

"Five steps. First, screen out clouds, because a cloud looks just like a dying crop. Second, describe each image with 16 numbers. Third, LOF gives each observation an anomaly score. Fourth, compare it with that tile's own threshold. Fifth, confirm an alarm only if two passes in a row are above it. The key idea is that every tile is judged against its own normal behaviour, not against one fixed number for everyone."

Key line: real stress lasts; a passing cloud does not.

Slide 6 · Detectors and decision rule ~55 s

"I compared three detectors. Isolation Forest separates points with random splits; LOF compares each point with its nearest neighbours; and the autoencoder is a small neural network that flags what it can't rebuild well. The hybrid averages all three. To turn a score into an alarm, each tile gets its own threshold from extreme value theory, set for a 3% false-alarm risk, which means only about 3 in 100 normal observations would cross it. The two-pass rule then confirms the alarm."

Slide 7 · Testing without ground truth ~60 s

"There is no ground truth, so I created it. I added synthetic stress to 15% of observations (sudden drops, slow declines and patchy damage) and measured how much the system finds. Because I added the stress myself, I know exactly where it is, so I can count what was caught and what was missed. I also tested different strengths, from 10 to 50 percent, and lengths, from one to three passes, over five random seeds. And I split the tiles into three groups: the detector learns on the first, the threshold is chosen on the second, and every number I report comes only from the third, which was never used for any decision."

Key line: every number comes from tiles the system never saw.

Slide 8 · Final result ~50 s

"On 832 unseen tiles: precision 0.876, recall 0.814 and F1 0.844. When it raises an alarm it's right about 88% of the time, and it finds about 81% of the stressed observations. There were only 350 false alarms in 20,800 observations, which is less than half a false alarm per tile over the whole season. The PR-AUC, which doesn't depend on any threshold, is 0.904, so the ranking itself is strong."

Key line: F1 0.844 on unseen tiles.

Slide 9 · Finding 1: combining detectors made it worse ~60 s

"The first surprise: the hybrid I set out to build was worse. LOF alone scored 0.844; the hybrid only 0.354. Its precision was fine, but recall fell from 0.81 to 0.23, so it went quiet. Isolation Forest alone scored only 0.179, so mixing it in pulled LOF down: averaging a strong detector with weak ones ruins the ranking. Stress is a local anomaly, unusual compared with similar observations, which is exactly what LOF measures. A large published benchmark also found LOF works best for local anomalies like these, so I kept LOF."

Key line: simpler won.

Slide 10 · Finding 2: the threshold mattered most ~60 s

"The biggest gain came from the threshold, not the model. Earlier, the threshold was computed on data that already contained the stress. With 15% stress injected, the top 15% of scores are the stress itself, so an 85th-percentile cut landed right on top of it, and recall was stuck near 0.6. Computing the threshold on clean data raised F1 from 0.693 to 0.844 with the same detector, and the confidence interval shows it isn't luck. I also replaced the plain percentile with extreme value theory, so the setting now has a meaning: a 3% false-alarm risk."

Key line: where you compute the threshold matters more than which model you use.

Slide 11 · Strengths ~45 s

"This map shows how well it detects stress of different strengths and lengths; darker is better. Whenever stress is 30% or more and lasts two passes or more, F1 is between 0.755 and 0.898, and for strong stress lasting three passes it reaches almost 0.9. It works for sudden drops and patchy damage. And it is 3.5 times better than simple hand-written rules, like a fixed percentage drop or a z-score; the best of those scored only 0.243."

Slide 12 · One tile, start to finish ~45 s

"Here's one test tile. The shaded bands are where stress was added; the dashed line is this tile's own threshold, and the crosses are confirmed alarms. All 8 stressed passes were caught, with no false alarms. The open circle crossed the threshold once but not on the next pass, so it was ignored. That is the two-pass rule removing a blip. This is the kind of output a user would get: which tile, and on which passes, to go and check."

Slide 13 · We turned the project into a research paper ~50 s

"We didn't stop at the project: we wrote it up as a research paper in IEEE conference format, with 10 figures, 7 tables and 28 references, and every number checked against the same results file as these slides. The title says it in short: 'ensemble dilution' means a weak model dilutes a strong one, and 'primacy of threshold calibration' means the threshold matters most. It makes three contributions: a clear negative result, that combining detectors made things worse; a practical insight, that where the threshold is computed matters more than the model; and a careful protocol where nothing is tuned on test data. It's drafted, and the next step is to submit it to a conference."

Key line: the project produced a research paper, not just a result.

Slide 14 · Conclusion ~35 s

"Three takeaways: a label-free detector that reaches F1 0.844 on unseen tiles; the simpler model beat the hybrid; and the threshold mattered more than the model. Overall, the project shows that careful testing and well-chosen simple methods can go further than complex ones. I'd like to thank Ms. Nandani Sharma for her guidance. Thank you, I'm happy to take questions."

Part 2 · Key terms in plain English

Term	What it means (say it like this)
Tile	A fixed square of farmland, 48 × 48 pixels at 10 m = 480 m × 480 m (about 23 hectares). It often contains more than one field. 3,328 of them.
Pass / observation	One satellite picture of one tile on one date. 25 kept per tile, so 83,200 in total.
MSAVI	Modified Soil-Adjusted Vegetation Index. Measures how green a tile is, like NDVI, but corrects for bright bare soil early in the season.
Unsupervised	Learns what "normal" looks like from the data itself, with no labelled examples of stress.
Anomaly score	A number saying how unusual an observation is. Higher = more unusual.
Isolation Forest	Splits the data randomly; points that get separated after only a few splits are unusual. Good at values that are extreme for the whole dataset.
LOF (Local Outlier Factor)	Compares how crowded a point's neighbourhood is with how crowded its 10 nearest neighbours' neighbourhoods are. Much emptier than its neighbours = anomaly. Score ≈ 1 is normal.
Autoencoder	A small neural network (16 → 32 → 4 → 32 → 16) that learns to rebuild normal data. A bad rebuild means an unusual observation.
Hybrid / ensemble	Averaging the normalised scores of several detectors. Here: Isolation Forest + LOF + autoencoder, equal weights.
Per-tile threshold	Each tile gets its own cut-off, learned from its own normal (clean) history.
EVT / GPD	Extreme value theory. Fit a Generalised Pareto curve to the top tail of a tile's clean scores, then put the cut-off where only q = 3% of normal observations would cross.
Persistence rule	Raise an alarm only when two consecutive passes are above the threshold.
Synthetic injection	Deliberately lowering the numbers of a known 15% of observations, so we know where the "stress" is and can score the system.
Fit / calibration / test	Three separate groups of tiles: learn normal (1,664) / choose the threshold and settings (832) / report results only (832).
Precision	Of all alarms raised, the fraction that were real stress. 0.876 ≈ 88 of 100 alarms are right.
Recall	Of all stressed observations, the fraction the system caught. 0.814 ≈ 81 of 100 caught.
F1	One score that balances precision and recall (their harmonic mean). 0.844.
PR-AUC	How well the scores rank stressed above healthy, over every possible threshold. 0.904.
False alarm	Stress flagged where none was injected. 350 out of 20,800 test observations.
Savitzky–Golay filter	A smoothing filter that removes one-off dips (like cloud shadow) from a time series but keeps real declines.

FORMULAS, IF ASKED

$$\mathrm{NDVI} = (\mathrm{NIR} - \mathrm{Red}) \,/\, (\mathrm{NIR} + \mathrm{Red})$$

$$\mathrm{MSAVI} = [2{\cdot}\mathrm{NIR} + 1 - \sqrt{((2{\cdot}\mathrm{NIR} + 1)^2 - 8{\cdot}(\mathrm{NIR} - \mathrm{Red}))}] \,/\, 2$$

$$\mathrm{F1} = 2 \cdot \mathrm{P} \cdot \mathrm{R} \,/\, (\mathrm{P} + \mathrm{R})$$

THE PIPELINE IN ONE BREATH

Screen clouds → 16 features per tile per pass → scale → LOF score → per-tile EVT threshold (3% risk) → alarm after 2 passes in a row.

Part 3 · Questions the panel may ask, and how to answer

Answers are short on purpose: say the first sentence, then add the number. ★ = must know. Amber = a trap or a likely follow-up.

A. The problem and the data

★ Why unsupervised and not a normal supervised model?

> Because labels don't exist at scale. A supervised model needs someone to visit fields and record which were stressed, which is slow and expensive. An unsupervised model needs only the images, so it can run anywhere Sentinel-2 covers.

Why Sentinel-2?

> It's free, has 10 m resolution, 13 spectral bands, and revisits every place about every five days. That is enough to follow one tile through a whole season.

★ Why MSAVI and not NDVI?

> Early in the season the ground is mostly bare soil, and NDVI gets confused by soil brightness. MSAVI has a built-in soil correction that adjusts itself, so it follows living plants instead of the soil.

What is a tile, and why tiles and not individual fields?

> A tile is a fixed 480 m square. The dataset comes as tiles, without field boundaries. A tile can contain several fields, so an alarm means “check this area”. Working at tile level also averages away pixel noise and gives one decision per tile per pass.

★ How do you handle clouds?

> Two ways. First, screening: an image is rejected if it is too uniform, has under 70% valid pixels, has too few distinct values, or has no texture (real fields have edges; cloud tops don't). Second, the two-pass rule: a cloud rarely sits on the same tile two passes in a row.

What are the 16 features, and why these?

> Per pass: mean, std, min, max, p25, median, p75, coefficient of variation, IQR and valid-pixel ratio. Over time: a smoothed median, a rolling median and rolling std, the change and percentage change from the last pass, and the difference from the median of all tiles. Together they describe level, spread and change, which is what stress affects. (Valid-pixel ratio is always 1.0 after screening, so effectively 15 features.)

Why only one region and one season?

> One full season over 3,328 tiles (83,200 observations) was enough to build and test the method properly. More regions and seasons are part of the future work. The side study also used the 2018 season.

B. Why this method

★ Explain LOF simply.

> For each observation, look at its 10 most similar observations and compare how crowded the neighbourhoods are. If this point sits in a much emptier area than its neighbours, it's unusual, so it gets a high score. A score near 1 is normal; well above 1 is an anomaly.

★ Why LOF and not Isolation Forest?

> Isolation Forest finds values that are extreme for the whole dataset. A stressed tile that is naturally low in vegetation isn't extreme overall; it's only unusual compared with similar observations. That local comparison is exactly what LOF does. Result: LOF 0.844 against 0.179 for Isolation Forest.

★ Why did the hybrid get worse? Isn't that a bug?

> No. I checked each detector alone: Isolation Forest scores only 0.179. Averaging LOF's strong ranking with a weak one produces a weak ranking, and the threshold then cuts in the wrong place. The hybrid's precision stayed about 0.77, but its recall fell to 0.230. ISO + LOF without the autoencoder gives the same 0.354, so Isolation Forest drives the damage. An earlier, separate stage of the project found the same thing.

Why not tune the hybrid's weights?

> The best weights would put everything on LOF, which is just LOF. Calling that a hybrid would be misleading.

Why not deep learning, like a CNN or LSTM?

> There are no labels to train a supervised network, and each tile has only 25 points per season, which is short for sequence models. I did test a neural network, the autoencoder, and it reached 0.528, below LOF. LOF is simpler, faster and runs on a CPU.

Why k = 10 neighbours?

> It was chosen on the calibration tiles, not the test tiles, and the result hardly depends on it: F1 stays between 0.805 and 0.844 for k from 5 to 75.

★ Why a separate threshold for each tile?

> Tiles behave differently: a rice paddy jumps up and down naturally, while a uniform field is calm. The calibrated thresholds differ by 4 times across tiles. One global cut-off would over-alarm the jumpy tiles and miss stress on the calm ones.

★ What is extreme value theory doing here?

> For each tile I take its clean scores, fit a curve (a Generalised Pareto distribution) to the highest ones, and set the threshold where only 3% of normal observations would cross. With only 25 scores per tile a plain percentile is crude, and the fitted tail gives a setting with a meaning: “3% false-alarm risk”.

Why 3%?

> It was chosen on the calibration tiles. It works like a dial: a lower risk gives fewer false alarms but more misses, and a higher risk gives earlier warnings but more false alarms.

★ Why two consecutive passes?

> Real stress lasts days to weeks, while a cloud or haze usually lasts one pass. The two-pass rule cut false alarms from 859 to 350 and raised F1 from 0.809 to 0.844. Three passes was too strict for two-pass events, and “2 out of 3” let in too many single blips (1,388 false alarms).

Why scale (standardise) the features?

> LOF works with distances. Without scaling, features with big numbers (MSAVI is stored in the thousands) would drown out small ones like ratios. The scaler is fitted only on the fit tiles.

CAREFUL: "DID YOU USE PCA?"

The early pipeline fits a PCA, but no detector uses its output, so PCA is **not** part of the method. If your resume says "PCA preprocessing", say: "PCA was in the early pipeline, but the final detector works directly on the 16 scaled features."

C. Evaluation and results

★ With no ground truth, how do you know it works?

I created labelled test data by adding known stress to clean data, then measured how much the system recovered. This is standard practice in anomaly-detection research. And every number comes from tiles that were never used for building or tuning.

★ Is synthetic stress realistic? (trap)

It's a controlled test, not the real world, and I say that openly. Its strength is that every method is compared on exactly the same data, so the comparisons are fair and reliable. Real stress is messier, so real accuracy will probably be lower. Checking against field records is the next step.

★ Explain precision, recall and F1 using your numbers.

Precision 0.876: of every 100 alarms, about 88 are real stress. Recall 0.814: of every 100 stressed observations, about 81 are caught. F1 0.844 balances the two.

Why F1 and not accuracy?

Only 15% of observations are stressed. A system that always says "healthy" would be 85% accurate and completely useless. F1 focuses on the stressed cases.

What do PR-AUC 0.904 and ROC-AUC 0.974 mean?

They measure how well the scores rank stressed observations above healthy ones, across every possible threshold. PR-AUC is the better one when positives are rare. They show the ranking itself is strong, not just a lucky threshold.

★ Are 350 false alarms a lot?

350 out of 20,800 observations is 1.7%, or about 0.4 false alarms per tile over the whole season. With precision 0.876, roughly 1 alarm in 8 is false. They also concentrate on a few noisy tiles: a third of tiles never had a false alarm.

★ Isn't changing the threshold just tuning on the test set? (trap)

No. The threshold rule, the 3% risk and k were all chosen on the calibration tiles; the test tiles were only used to report. I even kept the extreme-value rule although a percentile rule scored 0.001 higher on test (0.8455 against 0.8437), because picking by test score would be cheating.

★ How do you know the +0.151 F1 gain is real?

A bootstrap over 4,160 tile-and-seed combinations gives a 95% confidence interval of +0.142 to +0.159, far from zero, and a permutation test gives $p < 0.0005$. Every result is also averaged over five random seeds.

Isn't that gain just because of your 15% test setup? (trap)

Partly, and I'd say so. At a 15% stress rate the old rule's percentile lands right on the anomalies, so it is hurt most in this setting; with less stress it would lose less. But the principle holds everywhere: a threshold should be learned from clean data, not from data that already contains what you're trying to detect.

Does the model overfit? Does it generalise?

Fitting on all tiles versus only a separate half costs just 0.021 F1 (0.714 against 0.693), so it carries over to unseen tiles. If pushed: the test tiles are mixed in among the training tiles on the map, so testing on a whole new region would be a stronger check, which is future work.

How does it compare with simple rules anyone could write?

> I tested a fixed percentage drop, a per-tile z-score and a rolling-median rule, all through the same threshold and two-pass rule. The best rule scored 0.243 against 0.844 for LOF, which is 3.5 times better.

Where does it work best, and where is it weaker?

> It's strong for stress of 30% or more lasting two or more passes (F1 0.755–0.898). Very mild stress, 10–20%, is harder because it looks like normal day-to-day variation, and single-pass events are ignored on purpose because they could be clouds.

Why is slow (gradual) stress lower, at 0.614?

> The features compare each pass with a rolling baseline of recent passes. A slow decline drags that baseline down with it, so the difference stays small. More bands, like the water index NDMI, could catch it earlier.

What was the multispectral side study?

> A separate pipeline built from the raw 9-band data. Using four indices (MSAVI, NDVI, NDRE, NDMI) instead of one raised F1 from 0.243 to 0.351, a 44% gain. It used its own setup, so its numbers aren't comparable with the main 0.844. It also showed that last year's data is a poor baseline, because crop rotation changes what's normal.

D. Tough and practical questions

Can it tell drought from disease? (trap)

> No. It says “this tile is behaving unusually now”, not why. Finding the cause needs a field visit or extra bands (NDMI for water stress). That's the point of an early-warning screen: it tells you where to look.

Harvest causes a big drop. Won't it raise an alarm? (trap)

> Harvest happens on most tiles every year, so it is part of what LOF learns as normal from 1,664 training tiles; LOF flags what's unusual compared with similar observations. An unusually timed harvest could still trigger an alarm, which is a likely source of some false alarms. Crop calendars would reduce that.

Is it real-time? (trap)

> Not yet. It works as a season analysis: the smoothing step looks at the next two passes, and the two-pass rule waits one extra pass. A live version needs one-sided smoothing, which is in the future work.

★ What is the future scope? (not on the slides, so be ready)

> Frame it as building on a working system: add more spectral bands (a side study already showed +44% F1, and the water index NDMI should catch slow stress earlier), check the alarms against real field records, turn it into real-time alerts for each new satellite pass, and submit the research paper.

Would it work in India?

> In principle, yes: it needs no labels and Sentinel-2 covers India, so you would refit it on local tiles. Two adjustments: monsoon clouds (screening and the two-pass rule help) and smaller fields (use smaller tiles).

What tools did you use, and how long does it take to run?

> Python 3.14: NumPy, pandas, SciPy, rasterio for the satellite rasters, scikit-learn for Isolation Forest and LOF, PyTorch for the autoencoder, and Matplotlib for figures. The full study takes about 34 minutes the first time on a CPU, and later runs are faster because data is cached.

How is it reproducible?

> Fixed random seeds (five of them), and one command, run_all.py, rebuilds every table and figure. All the reported numbers are read from one results file, so the figures and the report can't disagree.

What was the hardest part? What did you personally do?

> [Make this true for you.] A good answer: “The hardest part was finding why recall was stuck near 0.6 whatever model I used. I traced it to the threshold being computed on data that already contained the anomalies, and fixing that gave the biggest improvement in the project.”

What would you do differently? What did you learn?

> I'd set up the three-way split and clean calibration from day one, get a little real ground truth early, and test on a held-out region. The lesson: a more complex model isn't automatically better; change one thing at a time, and check surprising results against the code.

★ **Tell us about your paper.**

> It's titled “Unsupervised Crop Stress Detection from Sentinel-2 Time Series: Ensemble Dilution and the Primacy of Threshold Calibration”, written in IEEE conference format with 10 figures, 7 tables and 28 references. “Ensemble dilution” means the hybrid got worse because a weak model dilutes a strong one; “primacy of threshold calibration” means where the threshold is computed matters most. Every number in it was checked against the same results file as the report and slides.

Is the paper published? Where will you submit it?

> It's drafted but not submitted yet. Say “drafted”, never “published”. The next step is to finalise it (author details, a final check of references) and submit it to a conference. [If your mentor has suggested a venue, name it here.]

DON'T SAY THESE

- **“Accuracy of 84%”**: say “F1 of 0.844”.
- **“It works in real time”**: it's a season analysis for now.
- **“PCA is part of my method”**: it isn't used by any detector.
- **“Tested on real crop stress”**: it was tested on synthetic stress.
- **“The hybrid is my final model”**: the final model is LOF alone.
- **“F1 0.718”**: that was an earlier stage; the final result is 0.844.
- **“The paper is published”**: it's drafted, not submitted.

Numbers sheet

Cover the right column and test yourself. ★ = must know.

What	Number
★ Tiles · size	**3,328 tiles · 480 m × 480 m (48 × 48 px at 10 m)**
★ Passes per tile · observations	**25 (2019 season) · 83,200**
Features	**16 (15 informative)**
★ Split: fit / calibration / test tiles	**1,664 / 832 / 832**
Synthetic stress	**15% of observations · 40% drop · 2 passes · 5 seeds**
★ Final precision / recall / F1	**0.876 / 0.814 / 0.844**
PR-AUC · ROC-AUC	**0.904 · 0.974**
★ False alarms	**350 of 20,800 (1.7%)**
★ LOF vs hybrid (F1)	**0.844 vs 0.354 (hybrid recall 0.230)**
Isolation Forest · autoencoder (F1)	**0.179 · 0.528**
★ Old threshold → new threshold (F1)	**0.693 → 0.844 (+0.151; 95% CI +0.142 to +0.159)**
Best hand-written rule (F1)	**0.243 (LOF is 3.5× better)**
Threshold settings	**q = 3% risk · k = 10 · ξ = −0.199**
Persistence: none vs two-pass	**F1 0.809 → 0.844 · false alarms 859 → 350**
Strong range (stress ≥30%, 2+ passes)	**F1 0.755 – 0.898**
Stress types: sudden / patchy / gradual	**0.844 / 0.809 / 0.614**
Cost of testing on unseen tiles	**0.021 F1 (0.714 vs 0.693)**
Side study: 1 index → 4 indices	**F1 0.243 → 0.351 (+44%)**
Thresholds vary across tiles by	**4×**
Seasonal swing of MSAVI	**about 2.7×**
Full run time	**about 34 minutes (CPU)**
★ Research paper (IEEE format, draft)	**10 figures · 7 tables · 28 references**

IF YOU BLANK OUT, COME BACK TO THIS

"No labels. LOF with a per-tile threshold and a two-pass rule. F1 0.844 on unseen tiles. Simpler beat the hybrid, and the threshold mattered more than the model."